WORKED SOLUTION

STEP II 1998, Question 6

Pure • Parametric equations

STEP II 1998, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s2-q6

The key idea

Both curves are cycloids, the paths traced by a point on the rim of a rolling circle, and everything about them gets simpler with the half-angle formulae $1+\cos\theta = 2\cos^2\frac{\theta}{2}$, $1-\cos\theta = 2\sin^2\frac{\theta}{2}$ and $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$. The gradients collapse to single tangents or cotangents of $\frac{\theta}{2}$, and the normal's equation shrinks to one tidy line. To show a line is a tangent to curve (2), find a point of (2) that lies on the line and where the curve's gradient equals the line's.

(The question's second curve has y_1 where it should say y_2; it's a misprint.)

Solution

Gradients

For curve (1), $\dfrac{\mathrm{d}x_1}{\mathrm{d}\theta} = 1+\cos\theta$ and $\dfrac{\mathrm{d}y_1}{\mathrm{d}\theta} = -\sin\theta$, so

$$\frac{\mathrm{d}y_1}{\mathrm{d}x_1} = \frac{-\sin\theta}{1+\cos\theta} = \frac{-2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\cos^2\frac{\theta}{2}} = -\tan\frac{\theta}{2},$$

valid wherever $\cos\frac{\theta}{2} \neq 0$, that is, $\theta \neq \pi$.

For curve (2), $\dfrac{\mathrm{d}x_2}{\mathrm{d}\theta} = 1-\cos\theta$ and $\dfrac{\mathrm{d}y_2}{\mathrm{d}\theta} = \sin\theta$, so

$$\frac{\mathrm{d}y_2}{\mathrm{d}x_2} = \frac{\sin\theta}{1-\cos\theta} = \frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\sin^2\frac{\theta}{2}} = \cot\frac{\theta}{2},$$

valid for $\theta \neq 0, 2\pi$.

At $\theta = \pi/2$, $\frac{\theta}{2} = \frac{\pi}{4}$: curve (1) has gradient -1 and curve (2) has gradient 1.

At $\theta = 3\pi/2$, $\frac{\theta}{2} = \frac{3\pi}{4}$, where tan $= -1$ and cot $= -1$: curve (1) has gradient 1 and curve (2) has gradient -1.

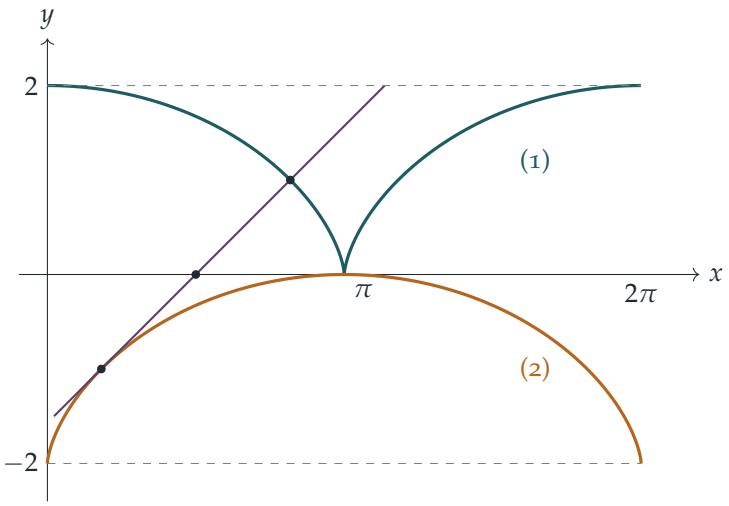
Sketch

Both x_1 and x_2 are non-decreasing in θ (their derivatives $1 \pm \cos\theta$ are never negative), so each curve runs left to right from $x = 0$ to $x = 2\pi$.

Curve (1) starts at $(0, 2)$ with gradient 0, falls to $(\pi, 0)$, then rises symmetrically to $(2\pi, 2)$. At $\theta = \pi$ both derivatives vanish and $|\mathrm{d}y_1/\mathrm{d}x_1| = |\tan\frac{\theta}{2}| \to \infty$, so there is a cusp at $(\pi, 0)$ with a vertical tangent. Since $0 \le y_1 \le 2$, the curve lies between $y = 0$ and $y = 2$.

Curve (2) has a cusp at $(0, -2)$ (vertical tangent, since $\cot\frac{\theta}{2} \to \infty$), rises to $(\pi, 0)$ where its gradient is $\cot\frac{\pi}{2} = 0$, then falls to the cusp $(2\pi, -2)$. It's an ordinary cycloid arch standing on the line $y = -2$.

The two curves touch at $(\pi, 0)$: curve (1) arrives there vertically, curve (2) passes over it horizontally. The picture is symmetric about $x = \pi$, since $\theta \mapsto 2\pi - \theta$ sends $x \mapsto 2\pi - x$ and leaves y unchanged on both curves. The plum line is the normal to (1) at $\theta = \pi/2$, which we meet below.



The normal to curve (1)

The tangent to (1) has gradient $-\tan\frac{\theta}{2}$, so the normal has gradient $\cot\frac{\theta}{2}$ (for $\theta \neq 0, \pi, 2\pi$; we deal with those below). The normal at the point $(\theta + \sin\theta,\ 1 + \cos\theta)$ is

$$y - (1 + \cos\theta) = \frac{\cos\frac{\theta}{2}}{\sin\frac{\theta}{2}}(x - \theta - \sin\theta).$$

Multiply through by $\sin\frac{\theta}{2}$ and use the half-angle forms:

$$y\sin\tfrac{\theta}{2} - 2\cos^2\tfrac{\theta}{2}\sin\tfrac{\theta}{2} = (x - \theta)\cos\tfrac{\theta}{2} - 2\sin\tfrac{\theta}{2}\cos^2\tfrac{\theta}{2}.$$

The last terms on each side are equal and cancel, leaving

$$\boxed{y\sin\tfrac{\theta}{2} = (x - \theta)\cos\tfrac{\theta}{2}}$$

as the equation of the normal.

This form also covers the exceptional values. At $\theta = 0$ it reads $x = 0$, the vertical line through $(0, 2)$, which is right because the tangent there is horizontal. At $\theta = 2\pi$ it gives $x = 2\pi$ in the same way. At $\theta = \pi$ it reads $y = 0$, the horizontal line through the cusp, perpendicular to the vertical cusp tangent.

Notice that the normal always passes through $(\theta, 0)$ on the x-axis. That's the point where the rolling circle touches the line it rolls on, a classic property of cycloids.

The normal is a tangent to curve (2)

Ask yourself Which point of curve (2) should the normal touch?

The tangent to (2) at parameter ϕ has gradient $\cot\frac{\phi}{2}$, and the normal has gradient $\cot\frac{\theta}{2}$. For ϕ, θ in $(0, 2\pi)$ these agree only when $\phi = \theta$, since cot is one-to-one on $(0, \pi)$. So the only candidate is the point of (2) with the same parameter θ.

That point is $(x_2, y_2) = (\theta - \sin\theta,\ -(1 + \cos\theta))$. Substituting into the normal's equation:

$$\text{LHS} = y_2 \sin\tfrac{\theta}{2} = -2\cos^2\tfrac{\theta}{2}\,\sin\tfrac{\theta}{2},$$
$$\text{RHS} = (x_2 - \theta)\cos\tfrac{\theta}{2} = -\sin\theta\cos\tfrac{\theta}{2} = -2\sin\tfrac{\theta}{2}\cos^2\tfrac{\theta}{2}.$$

These are equal, so the normal to (1) at θ passes through the point of (2) with parameter θ. At that point the gradient of (2) is $\cot\frac{\theta}{2}$, the same as the normal's. A line through a point of a curve with the curve's gradient there is the tangent at that point, so the normal to (1) is a tangent to (2), as required.

The special values work too. For $\theta = \pi$ the normal $y = 0$ touches (2) at its top $(\pi, 0)$, where the gradient of (2) is 0. For $\theta = 0$ (or 2π) the normal $x = 0$ (or $x = 2\pi$) is the vertical tangent at the cusp of (2).

Checks

- At $\theta = \pi/2$ the normal is $y = x - \frac{\pi}{2}$. It passes through $(\frac{\pi}{2} + 1, 1)$ on (1), through $(\frac{\pi}{2}, 0)$, and through $(\frac{\pi}{2} - 1, -1)$ on (2), where (2) has gradient 1, as found in the first part. That's the plum line in the sketch.
- A numerical check of the gradients by finite differences agreed with $-1, 1$ and $1, -1$ to six decimal places. At 1000 random values of θ, the normal passed through both points and was parallel to the tangent of (2), to within 10^{-12}.

Common slips

- Leaving the gradients as $-\sin\theta/(1 + \cos\theta)$ and then grinding through messy algebra for the normal. The half-angle forms make the normal a two-line calculation.

- Showing only that the normal meets curve (2). Meeting isn't enough: you also need the gradients to match at the meeting point.

Beyond STEP

The curve traced out by the centres of curvature of a curve is its *evolute*, and it is also the envelope of the curve's normals. This question shows that the evolute of a cycloid is an identical cycloid shifted by half an arch. The distance along the normal from (1) to its point of contact with (2) is $4|\cos\frac{\theta}{2}|$, which is exactly the arc length of (2) from that point to its top. So (1) is traced by the end of a string unwinding from (2). Huygens used this to build a pendulum whose period doesn't depend on its amplitude. For curvature, evolutes and involutes, try Pressley, *Elementary Differential Geometry*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.